

Client: Matthew Morelli
Company: NRG Efficient Homes
Project:

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Thermal Performance Computation

This document verifies the Total Thermal Performance R-value of the building profile details below. These results have been calculated following the appropriate procedures outlined in AS/NZS 4859:2:2018, with bridging allowances using NZS 4214:2006 and in accordance with Volume One NCC 2019.

Application profile: Concrete Wall

Thickness	Material Description	Emittance		R-value (m²K/W)	
		Out	In	Summer	Winter
	Outdoor air film			0.040	0.030
150.0mm	SOLID CONCRETE (2400 Kg/m³)			0.104	0.104
25.0mm	KOOLTHERM® K12 (R-value = 1.09/1.09m²K/W (in/out)) 28mm Metal Furring Channel - @ 450mm ctrs (7.8%)			0.417	0.426
13.0mm	GYPSUM BOARD			0.076	0.076
	Indoor air film			0.120	0.120
Total R-value (m²K/W)				0.8	0.8
Overall U-value (W/m²K)				1.32	1.32

Notes:	Summer	Winter
Indoor temperature	24.0 °C	18.0 °C
Outdoor temperature	36.0 °C	12.0 °C
Temperature difference	12.0 K	6.0 K

This Calculation is valid for the products specified above only.


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